

Thiago de Paula Oliveira

Statistician | Statistical Computing | Agriculture | Genetics

LOCATION Edinburgh, UK

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EXECUTIVE PROFILE

Statistician and analytics developer with 14+ years of experience converting advanced modelling into practical tools for decision-making in agriculture, quantitative genetics, and applied health. Combines mixed-model, Bayesian, and longitudinal methods with production-ready engineering in R, C++, SQL, Bash, Shiny, Docker, and Git to deliver reliable, stakeholder-ready outputs.

CAREER HIGHLIGHTS

- **14+ years** delivering applied statistical solutions across commercial and research settings.
- **2 CRAN packages** developed and maintained for correlation and longitudinal concordance workflows.
- **3 domains** of delivery: agriculture, quantitative genetics, and applied health analytics.

CORE COMPETENCIES

STATISTICAL METHODS	Mixed models, Bayesian analysis, multilevel modelling, longitudinal analysis, simulation, experimental design
DATA AND SOFTWARE	R, C++, SQL, Bash, Shiny, Docker, CI, Git, reproducible pipelines, package development
DOMAIN FOCUS	Quantitative genetics, breeding-program analytics, sustainability indices, athlete and health-monitoring analytics
DELIVERY STYLE	Decision-support tooling, technical reporting, stakeholder communication, mentoring and code review

PROFESSIONAL EXPERIENCE

APR 2023 – PRESENT	Consultant Statistician AbacusBio • Lead statistical delivery for livestock, crop, and agri-tech programmes, building decision-support workflows for genetic evaluation, genomic QC, and trial analysis. • Designed economic and sustainability selection indices to align breeding, operational, and commercial objectives with transparent ranking logic. • Built production workflows in R, C++, SQL, and Bash with Docker and Git, improving reproducibility, handover quality, and deployment readiness. • Delivered Shiny applications, technical reports, and stakeholder summaries for scientists, breeders, and commercial teams. • Mentored colleagues through code review and guidance on modelling strategy, package standards, and reproducible development.
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JAN 2020 – MAR 2023

Research Fellow in Quantitative Genetics

[The Roslin Institute, University of Edinburgh](#)

- Awarded a Marie Skłodowska-Curie COFUND fellowship focused on quantitative-genetic methods for sustainability-oriented breeding programmes.
- Developed mixed-model, Bayesian, and simulation approaches for plant-breeding data, selection strategy, and programme monitoring.
- Produced methods and analytical workflows for partitioning trends in genetic mean and additive genetic variance.
- Collaborated with genetics, breeding, and computing teams to translate methodology into implementable analytical pipelines.

JAN 2019 – MAR 2023

Postdoctoral Researcher in Statistics

[Insight Centre for Data Analytics, University of Galway](#)

- Developed Bayesian, multilevel, and longitudinal models for athlete readiness, fatigue, menstrual-cycle dynamics, and health-monitoring programmes.
- Built forecasting and dashboard pipelines for COVID-19 and related public-health decision support.
- Worked with clinicians and sports scientists to convert modelling outputs into practical tools and reporting workflows.

2019

Statistical Consultant

[Orreco](#)

- Developed statistical models and dashboard outputs for elite-athlete monitoring, recovery, and performance support using biomarker, GPS, and wellness data.

2017 – 2019

Lecturer in Statistics

[University of São Paulo \(ESALQ/USP\)](#)

- Taught experimental design, calculus, linear models, and statistical programming at undergraduate and postgraduate levels.
- Supervised applied projects connecting statistics with agriculture and genetics, with emphasis on reproducible workflows.

■ PRODUCTS AND OPEN-SOURCE CONTRIBUTIONS

- **matrixCorr** — R package for correlation and agreement estimation in a unified framework for method comparison and applied reporting. [[GitHub](#); [CRAN](#)]
- **lcc** — CRAN package for longitudinal concordance and accuracy profiling using polynomial mixed-effects models. [[GitHub](#); [CRAN](#)]
- **Selected contributions** — Contributions to [AlphaSimR](#), [AGHmatrix](#), [AlphaPart](#), and [Reads2Map Tools](#) in breeding simulation, matrix methods, and genotyping workflows. [[GitHub](#); [GitHub](#); [GitHub](#); [GitHub](#)]

■ EDUCATION

2014 – 2018

PhD in Statistics

[University of São Paulo \(ESALQ/USP\)](#)

Dissertation on longitudinal concordance analysis using polynomial mixed-effects models. Advisors: Silvio Sandoval Zocchi and John Hinde. [[PDF](#)]

2016	Visiting Research Internship University of Galway Research placement in concordance-analysis methodology under the supervision of John Hinde.
2012 – 2014	MSc in Statistics University of São Paulo (ESALQ/USP) Research in mixed-effects models for longitudinal fruit-colour measurement.
2007 – 2012	BSc in Agricultural Engineering University of São Paulo (ESALQ/USP) Training in agricultural systems, experimentation, and quantitative analysis.

■ SELECTED PUBLICATIONS

- Richardson, C., Amer, P., Post, M., **Oliveira, T. d. P.**, et al. (2025). [Breeding for sustainability: development of an index to reduce greenhouse gas in dairy cattle](#). *Animal*.
- Oliveira, T. d. P.** & Newell, J. (2024). [A hierarchical approach for evaluating athlete performance with an application in elite basketball](#). *Scientific Reports*.
- Oliveira, T. d. P.**, Obsteter, J., Pocrnic, I., Heslot, N. & Gorjanc, G. (2023). [A method for partitioning trends in genetic mean and variance to understand breeding practices](#). *Genetics Selection Evolution*.
- Oliveira, T. d. P.**, Bruinvels, G., Pedlar, C., Moore, B. & Newell, J. (2021). [Modelling menstrual cycle length in athletes using state-space models](#). *Scientific Reports*.

■ AWARDS

- 2020 — [Marie Skłodowska-Curie COFUND Fellowship](#), quantitative genetics and genomics of plant breeding.
- 2010 — Honorable Mention, 18th USP International Symposium of Undergraduate Research, University of São Paulo.

■ LANGUAGES

- Portuguese (native), English (professional proficiency), Spanish (reading/listening).